Project Proposal: Sign Language Learning Website

1. Introduction

- **SDG Concept: SDG 4 and SDG 10**

SDG 4 (Quality Education): The goal of this SDG is to "ensure inclusive and quality education for all and promote lifelong learning opportunities for all." One way that a sign language learning website can contribute to this sustainability goal is that it can provide resources for people to learn sign language at their own pace.

SDG 10 (Reduced Inequalities): The objective of this SDG is to "reduce inequality in all its forms and promote inclusion." One way that a sign language learning website can contribute to this sustainability goal is that it can facilitate communication between deaf or hard of hearing individuals and those who are not familiar with sign language.

- **Gaps / Problems**

Many individuals are keen to learn sign language; however, few have an opportunity to take sign language classes, receive guidance from certified instructors, or access relevant resources. Additionally, many learners at the beginner's level may lack clarity on how to start their studies or ensure that their movement follows standard sign language grammar and rules. These barriers contribute to the demand-supply gap in sign language learning by limiting accessibility to the required resources and causing frustrations among individuals who want to be involved in the communication methods used by the deaf community.

- **Manual Process / Existing Process**

Currently, learners can acquire sign language through various methods such as attending classroom instruction, reading books or watching videos, participating in social media challenges, or simply asking a friend who can sign. Although these practices may work to differing degrees, they may also provide a fragmented learning experience that lacks cohesive lesson planning, practice opportunities, and progress tracking.

- **Solution / Possible Technology**

I think it is possible to develop a web-based sign language learning system which will arrange the material in lessons/categories. The program could be created using HTML/CSS/JavaScript and a database. The system would allow one to complete interactive lessons, quizzes, practice activities, track progress, and more.

- **Proposed System: Sign Language Learning Website**

The proposed Sign Language Learning Website will offer a learning platform where users will be able to acquire knowledge of the basics of this language. The website will have categories from which users will be able to learn, watch demonstrations, take quizzes, and track their progress. It will also facilitate learning American Sign Language, making it more accessible and convenient for users.

2. Objectives

- **Objective**

The project will aim to:

• Develop an interactive web page through which it will be possible to learn the basics of sign language in a new and engaging way.

• Provide visual feedback based on web camera indications about the correctness of performed signs in real-time.

• Make the page interesting and customizable to increase the motivation of the users by providing interactive lessons with quizzes.

• Raise awareness among users about the importance of sign language for promoting inclusive communication among people with different physical conditions.

• Ensure equal opportunities by providing easily accessible educational content.

3. Possible System Features / Work Breakdown Structure

Sign Language Learning Website

- **1. User Management**
  - 1.1 User Registration
  - 1.2 User Login
  - 1.3 User Profile
  - 1.4 Logout
- **2. Learning Materials**
  - 2.1 Sign Language Alphabet
  - 2.2 Numbers
  - 2.3 Common Greetings
  - 2.4 Common Words and Phrases
  - 2.5 Family and People
  - 2.6 School-Related Signs
  - 2.7 Everyday Activities
  - 2.8 Images / Video Demonstrations
- **3. Learning Activities**
  - 3.1 Lesson Selection
  - 3.2 Sign Identification
  - 3.3 Multiple-Choice Questions
  - 3.4 Matching Activities
  - 3.5 Practice Exercises
- **4. Assessment**
  - 4.1 Quizzes
  - 4.2 Score Calculation
  - 4.3 Correct/Incorrect Feedback
  - 4.4 Quiz Results
- **5. Progress Tracking**
  - 5.1 Completed Lessons
  - 5.2 Quiz Scores
  - 5.3 Learning Progress
  - 5.4 Achievement/Completion Indicators
- **6. Admin Management**
  - 6.1 Admin Login
  - 6.2 Add Lessons
  - 6.3 Edit Lessons
  - 6.4 Delete Lessons
  - 6.5 Manage Quiz Questions
  - 6.6 Manage Users